

AI-Powered Fraud Detection System for Banking Transactions

Project Summary

This project introduces an advanced fraud detection framework for banking transactions, leveraging machine learning (ML) and generative AI to identify and analyze suspicious behavior. The core system combines unsupervised anomaly detection with behavioral risk scoring and generates AI-driven explanations to support fraud analysts. The primary goal is to achieve early detection of fraudulent patterns and provide clear, explainable insights for timely decision-making.

The Challenge

Modern digital banking systems handle enormous volumes of daily transactions. Fraud often manifests as subtle, unusual patterns rather than simple violations of predefined rules. The project addresses this by designing a system for continuous transaction monitoring, utilizing ML techniques for anomaly detection, and generating contextual explanations to streamline fraud investigations.

Methodology and Architecture

The fraud detection framework operates through several key analytical stages:

- **Data Processing:** Ingestion and thorough preprocessing of banking transaction data.
- **Feature Engineering:** Creation of features to capture critical behavioral signals, such as transaction hour and deviation from typical transaction amounts.
- **Anomaly Detection:** Use of the **Isolation Forest** algorithm as the central model. This algorithm is highly effective because fraudulent activities are rare and distinct, requiring fewer splits to isolate them from normal data.
- **Risk Scoring:** Application of rule-based logic for fraud risk scoring, based on detected anomalies and behavioral indicators.
- **AI Explanations:** Generation of descriptions by a generative AI model detailing *why* a specific transaction is flagged as suspicious.
- **Investigation Support:** Provision of interactive dashboard visualizations to aid fraud analysts in their investigation process.

Key Features Monitored by the ML Model:

- Transaction Amount
- Transaction Hour
- Login Attempts
- Account Balance
- Transaction Amount Z-Score
- Balance to Transaction Ratio

Project Results

Metric	Value
Total Transactions Monitored	2512
Anomalies Detected	126
High-Risk Transactions Flagged	61
Fraud Detection Rate	2.43%

Fraud Monitoring Dashboard

An interactive dashboard, built with Python visualization libraries, was developed to assist analysts. It offers views of fraud risk distribution, transaction behavioral patterns, and lists of high-risk transactions. Analysts can use interactive tools to inspect specific transactions and review the AI-generated explanations for the suspicious activity.

Business Value

The implemented system demonstrates the significant role of machine learning and AI in enhancing banking risk management. By ensuring early anomaly identification and providing explainable insights, the system empowers fraud analysts to prioritize their investigations effectively, reduce manual effort, and ultimately strengthen the bank's financial security posture.

Future Scope

Potential future enhancements include integrating real-time streaming pipelines for live monitoring, exploring deep learning models for advanced behavioral profiling, developing automated alerting systems, and deploying the solution as a robust web-based dashboard using frameworks like Streamlit or Flask.